Case Study: *C. trachomatis* — Diagnostic Target Selection and In-Silico Oligonucleotide Screening

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Executive summary

This project computationally verified a panel of diagnostic candidate targets for a multiplex qPCR/TaqMan assay for *Chlamydia trachomatis*. Four chromosomal loci (mutL, aroB, PSD2, and group_306) were retained as candidate targets, represented by seven independently screened oligo sets, alongside two candidate windows on the organism's multicopy diagnostic plasmid. The proposed duplex assay combines one chromosomal locus with the plasmid target, providing an independent genomic marker alongside the multicopy target, which has the potential to improve analytical sensitivity.

Candidate sets were evaluated using a common acceptance framework covering calculated primer/probe Tm and Tm gap, standardized BLAST specificity against the human genome and nucleotide databases, pair-level off-target analysis, and three-way structural assessment of all primer–probe interactions. All nine retained sets passed the defined in-silico acceptance criteria. The work is computational: amplification efficiency, analytical sensitivity, specificity in real samples and multiplex performance remain to be established experimentally.

A central feature of the workflow is independent verification of intermediate results. Examination of genome and plasmid alignments, direct measurement of inherited assumptions such as the plasmid's GC content, and independent assessment of candidate-ranking criteria identified several issues that would otherwise have propagated into the final design. The resulting workflow therefore combines target discovery with explicit QC gates and traceable decision criteria.

Background and objectives

Chlamydia trachomatis is an important target for nucleic-acid amplification testing, where assay performance depends on selecting sequences that are both conserved across the target population and sufficiently differentiated from related organisms and human DNA. A duplex design combining a conserved chromosomal locus with the organism's multicopy diagnostic plasmid provides two independent target classes: a stable genomic marker and a higher-copy target with the potential to improve analytical sensitivity.

The objective of this project was computational: to identify conserved candidate regions and rigorously evaluate primer/probe sets for inclusivity, exclusivity, thermodynamic suitability and structural compatibility before experimental testing. Two structurally different target classes were pursued: four chromosomal genes (mutL, aroB, PSD2, group_306) and the species' multicopy cryptic plasmid, each needing its own genome curation, consensus-building and design pipeline.

Methodology: from raw genomes to a candidate panel

The 125 RefSeq genomes used throughout this pipeline are the result of a provenance audit: the project's original genome inventory also included 149 GenBank-only complete genomes, which a BioProject-level audit traced to a single study of deliberately engineered laboratory strains — transposon-mutant and antibiotic-resistance-tagged constructs — rather than natural clinical isolates, and excluded outright. The 125 RefSeq genomes are the clean baseline this pipeline builds from. Even so, none of the 125 were accepted on RefSeq status alone — each was independently re-verified and tiered into primary, quality-review, or excluded sets, as the Genome curation and Provenance audit rows below show.

Stage	Genome count	What it means
Provenance audit (bin/02)	125	Full RefSeq set, tiered: 97 primary / 10 quality-review / 18 excluded
Prokka annotation (bin/05)	107	Primary + quality-review, annotated
Panaroo pangenome (bin/06)	97	Primary tier only (904 gene clusters, 874 core); quality-review genomes are not pangenome inputs
Consensus-building / target sequence (bin/08–09)	97	The same 97 primary-tier genomes build the consensus each candidate target's sequence is drawn from

107 is the count that enters annotation; 97 is the count that drives everything after it — the pangenome, each target's consensus sequence, and the inclusivity claim itself. Quality-review genomes are annotated so they can still be examined later, but they are kept out of the pangenome and the consensus rather than allowed to shape them. They are held back as an independent test of the finished oligos (see Final results). The 18 excluded genomes never enter any of this past the provenance audit.

The ten quality-review genomes split into three distinct flags, not one generic concern. Five were flagged on raw assembly metrics measured against the panel: two carry a CheckM contamination score of 7.15% and a chromosome length roughly 41,100 nt longer than the panel median of 1,042,743 bp; one is 19,111 nt longer with no contamination flag; and two others are 16,904 nt and 9,511 nt shorter, with CheckM completeness scores of 91.61% and 93.78% respectively (completeness below 100% plus a shorter-than-median length both point the same way: a genuinely incomplete assembly, not a sequencing artifact). Four more — all from a single BioProject whose own title records "Laboratory adaptation of *Chlamydia trachomatis*" — are genuine clinical isolates that had been serially passaged in culture, which can introduce drift even though the original infection was natural. The tenth carries no flagged assembly metric at all; its BioProject metadata names a different organism entirely (*Bordetella pertussis*) against a *C. trachomatis*-labeled strain — a submission-metadata inconsistency flagged for exactly that reason, not a sequence-quality one.

The chromosomal pipeline ran as a sequence of purpose-built scripts, each producing the input the next stage needed:

Stage	Script	What it produced
Genome curation	bin/01	125 RefSeq *C. trachomatis* genomes downloaded and QC'd
Provenance audit	bin/02	Tiering: 97 primary, 10 quality-review, 18 excluded
Species + redundancy check	bin/03	fastANI: all pairs ≥98.90% identity, supporting species-level consistency: every genome is consistent with *C. trachomatis*
ompA serovar typing	bin/04	14 serovar groups identified
Annotation	bin/05	Prokka gene calls across 107 genomes
Pangenome	bin/06	Panaroo: 904 gene clusters, 874 core
Exclusivity screening	bin/07	692 of 874 core genes with a gap of at least 150 bp below 85% identity to every genome in the 123-genome exclusivity panel
Candidate region + consensus	bin/08–09	A majority-consensus sequence per candidate region
Primer3 design + QC	run manually against the consensus windows	Oligo sequences, real-computed Tm, structure
BLAST specificity screen	manual web BLAST	Specificity against the human genome and core_nt

The plasmid pipeline (bin/12–17, added later in the project) followed the same shape from a different starting point: 34 of the 125 genomes turned out to already carry an assembled ~7.5kb second contig — the species' cryptic plasmid — so no fresh genome curation was needed, only extraction, orientation correction, consensus-building and design against that real data.

Both pipelines converge on the same acceptance standard, covered next.

Standardized oligo acceptance framework

Oligos were designed with primer3-py inside the conserved consensus windows identified by the genome analysis, and every candidate was then screened against one acceptance framework. The standards below are recorded in the project's decision log (D6 and D18–D21) and were applied in the same way to every candidate set, so that results are directly comparable across genes and across both target classes. Each standard is set out with the reason it is there.

Requirement	Standard applied	Why it matters
Tm / Tm gap	primer3-py `calc_tm`, computed on the real candidate sequence; ≥5°C Tm gap hard floor, ≥7°C preferred (D6, D19)	The probe has to be bound before the primers extend, so a minimum gap between probe and primer Tm is a design requirement. Calculating every Tm with one tool on the actual sequence keeps values comparable across sets.
BLAST parameters	Fixed project-wide: Expect=1000, word_size=7, filter off, against the human genome and core_nt (excluding the *C. trachomatis* taxid) (D18)	Oligos this short only return weak, partial matches under permissive settings. Fixing the parameters means a "clean" result means the same thing for every set.
Off-target logic	A hit counts as a predicted amplicon only if the forward and reverse primers both bind nearby on the same accession, correctly oriented. A probe plus one primer alone is not a predicted amplicon (D20)	Amplification needs both primers. Judging at the level of the pair avoids rejecting good sets for single-oligo matches that cannot produce a product.
Structural check	primer3-py `calc_hairpin`, `calc_homodimer` and `calc_heterodimer` on all three pairs (F-R, F-Probe, R-Probe), severity set by the structure's own melting Tm (D21)	A TaqMan reaction contains three oligos that can all interact. Checking only the forward–reverse pair leaves two of the three possible interactions untested.
Aggregate scoring	No blended score. Tm gap, BLAST cleanliness and structure are each reported and must each pass on their own	A single combined number can hide a failed criterion behind passing ones, so each criterion is judged separately.

The same five checks — real Tm, standardized BLAST, pair-level logic, three-way structure, no blended score — were then applied, unchanged, to the second target class.

Extending the method: the plasmid target

A preliminary plan proposed curating a separate collection of plasmid records. Inspection of the existing genome dataset identified a more direct route: 34 of the 125 RefSeq assemblies already contained an assembled ~7.5kb second contig (7,415–7,510bp across all 34) corresponding to the species' cryptic plasmid. That existing, already-verified starting point replaced the proposed fresh curation entirely.

Aligning the 34 raw contigs with MAFFT produced an alignment roughly twice the length of the input sequences — a strong signal, given how closely length-matched the inputs were, that something other than ordinary divergence was going on. The cause was RefSeq's own submission history: a circular plasmid gets linearized at an arbitrary point, and different submissions started (and in some cases

oriented) it differently. Six anchor probes against a reference genome were used to detect each sequence's rotation and strand, correcting 31 of 34 (3 were flagged as low-confidence and excluded — the same 3 that an earlier, unrelated chromosomal QC pass had already flagged independently). Re-aligning the corrected 31 gave a 7,606bp alignment, in line with the raw sequence length.

A majority-rule consensus from the 27 primary-tier genomes came out 98.3% identical across all of them, with the remaining 1.65% correctly N-masked as real cross-strain variation. That consensus also settled a specific factual claim from the original handover — that the plasmid runs about 50% GC, higher than the ~43% chromosomal figure, which had been used to justify a higher probe-Tm target. The measured consensus GC is 35.4%, *lower* than the chromosome, not higher, so the design used the same Tm settings as the chromosomal panel instead. Primer3 design against the two longest N-free windows of the consensus (4,860–5,277 and 1,315–1,612) then went through the identical D6–D21 standard as every chromosomal set. These 34 plasmid sequences come from the curated genome collection, not from a survey of plasmid diversity, so conservation is shown across the sequences examined only. The plasmid design is a standard consensus design; the nvCT variant, a documented plasmid deletion, is not covered, because its published coordinates could not be independently confirmed (see Scope and limitations).

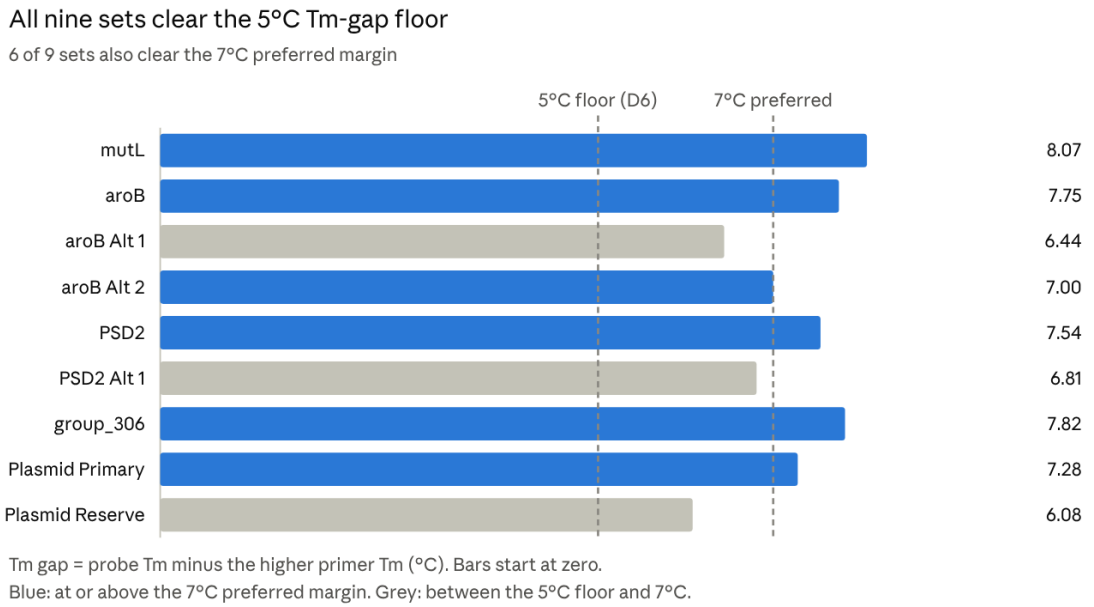
Verification checkpoints in practice: three issues caught before candidate selection

Three concrete errors were caught during the plasmid work by refusing to treat an intermediate result as correct just because it ran without error:

Issue	What it would have caused, unnoticed	How it was caught	Fix
Circular-genome alignment doubling	A consensus built from a 13,075bp alignment of ~7,500bp sequences — largely spurious gap columns, not real biology	The alignment length itself was implausible for sequences that closely length-matched	Anchor-probe rotation/strand correction before alignment; the corrected re-alignment (7,606bp) confirmed the fix
Plasmid candidate ranking rule	A Tm-gap/length-only ranking would have locked in a candidate with a flagged MEDIUM-severity primer hairpin over an equally viable, fully clean alternative — in both design windows	Computing the full three-way dimer check for every surviving candidate, not only the nominal top pick	Re-ranked by structural cleanliness first, then Tm gap and probe length (logged as decision D24)
Plasmid GC% claim	A higher, incorrect probe-Tm target, adopted on the strength of an unverified "~50% GC" claim	Measuring the actual consensus GC directly (35.4%) instead of trusting the figure	Kept the same Tm settings already verified on the chromosomal panel

In each case the error was invisible from the output of the step that produced it — it only showed up once that output was checked against an independent computation or a plausibility test, rather than passed forward as given.

Final results: the in-silico candidate panel



computed with primer3-py · 9 candidate sets

The nine sets break down as seven chromosomal sets across four genes — one for mutL, three independent loci for aroB (primary, Alt 1, Alt 2), two for PSD2 (primary, Alt 1), and one for group_306 — plus two candidate windows on the plasmid (Primary, Reserve).

The panel's nine sets are computationally verified diagnostic candidates, not a single nine-oligo reaction: the proposed duplex assay deploys one chromosomal locus alongside the plasmid target, with the rest held as backups. Every locked set clears the Tm-gap floor, and the two structural/specificity checks below hold for all nine — no plausible off-target amplicon is predicted in the human genome or against non-target *Chlamydia* species for any set, under the defined screening conditions (BLAST parameters fixed project-wide; a predicted amplicon requires both primers).

Target / set	Panel	Structure (three-way dimer check)	BLAST specificity
mutL	Chromosomal panel	Clean, LOW max severity	Clean; pair-level check clear
aroB (primary)	Chromosomal panel	Clean, LOW max	Clean pair-level; shares a conserved locus with *C. suis* at forward primer +

Target / set	Panel	Structure (three-way dimer check)	BLAST specificity
		severity	probe only — no completing reverse-primer hit, so no plausible amplicon
aroB Alt 1	Chromosomal panel	LOW	Clean
aroB Alt 2	Chromosomal panel	LOW	Clean
PSD2 (primary)	Chromosomal panel	LOW	Clean
PSD2 Alt 1	Chromosomal panel	LOW	Clean
group_306	Chromosomal panel	Clean — no hairpin on any of the three oligos	Clean
Plasmid Primary	Plasmid target	0 flags	Clean; matches 7 published *C. trachomatis* lab cloning vectors at 100% identity (expected: these are native-plasmid-derived laboratory reagents, not expected in clinical samples other than as laboratory carry-over); weak partial *C. muridarum* hit on the probe only, no completing pair
Plasmid Reserve	Plasmid target	0 flags	Clean; same lab-vector matches, no non-target *Chlamydia* hits

Cross-interactions between the chromosomal and plasmid oligos of the proposed duplex were also checked (`bin/19`, decision D29). For all 14 possible chromosomal-plus-plasmid pairings (nine cross pairs each), no cross-dimer reaches MEDIUM or HIGH severity on the scale used within each set; the strongest structure Tm is 21.5 °C under default conditions, or 27.0 °C under the project's reaction conditions. This is an oligo–oligo thermodynamic check only: BLAST cross-hits between the targets, detection-channel assignment and real multiplex behaviour remain for laboratory evaluation.

Check against the quality-review genomes. Because the ten quality-review genomes were kept out of the consensus, they also serve as an independent test of the finished design. On the chromosomal panel, each of the ten was tested directly against each of the seven finished chromosomal oligo sets (bin/18, BLAST of all 21 oligos), giving 70 genome-by-set cases, with the 97 primary genomes run against the same sets as a control. Of the 70 cases, 69 bound exactly, one (a probe with a single internal mismatch in a backup set) bound within tolerance, and none was at risk or failed. On the plasmid target,

the quality-review genomes that carry a plasmid were aligned alongside the primary set and checked against the consensus: all sit at 99.95–100% identity to it.

Lessons for computational assay design

- **Recompute inherited metrics rather than propagating reported values.** A Tm, a GC% or a specificity figure taken from a handover note or the literature is a claim, not a fact, until it's regenerated from the actual sequence with the actual tool this project relies on.
- **Judge each criterion on its own rather than blending them into one score.** A single aggregate number can hide the one failure mode it doesn't model. Tm gap, BLAST cleanliness and structure are each reported and must each pass independently, so a strength in one cannot offset a failure in another.
- **An implausible intermediate result is worth stopping for.** A doubled alignment length, a suspiciously round percentage, a candidate that scores identically to another — each is a cheap, early signal that something upstream needs a second look, before it shapes every downstream decision.
- **Check every relevant combination, not just the obvious one.** A three-oligo TaqMan set has three pairwise interactions (F-R, F-Probe, R-Probe); checking only F-R because it's the most familiar leaves two-thirds of the real risk unexamined.
- **Verify quantitative assumptions before allowing them to drive design decisions.** The plasmid's claimed GC% and a proposed Tm target built on it were both wrong, and the fifteen minutes it took to measure the real consensus GC was cheaper than designing against the wrong number.

Scope and limitations

- **Computational scope.** Every result here is a prediction from sequence data and software. Amplification efficiency, analytical sensitivity, specificity in real samples and multiplex performance have not been measured; they are the purpose of the experimental stage.
- **Reference population.** Inclusivity is shown for the 97 primary-tier genomes drawn from 125 RefSeq complete assemblies; lineages absent from RefSeq complete genomes are not represented. The plasmid evidence comes from the 34 plasmid-carrying assemblies in that collection, not from a survey of plasmid diversity.
- **nvCT variant.** The plasmid design is a standard consensus design. The nvCT variant, which carries a deletion in the plasmid, is not claimed to be covered: its published coordinates and accession numbers could not be independently confirmed, so variant-specific design was set aside rather than guessed. The chromosomal locus in the proposed duplex is the intended safeguard against this.
- **Specificity statements.** "Clean" and "no predicted off-target amplicon" describe results under the defined screening conditions (BLAST parameters, databases and the pair-level rule above). They are not a measurement of biological cross-reactivity.
- **The duplex is a proposal.** A second marker is a potential safeguard against false negatives, contingent on inclusivity, primer performance and the behaviour of both targets in one reaction. Oligo–oligo dimer interactions between the two sets were checked; BLAST cross-hits between targets, dye assignment and real multiplex behaviour were not.

Conclusion

The in-silico design phase produced five candidate targets represented by nine primer/probe sets and 27 oligos, all passing the same defined thermodynamic, structural and specificity criteria. The workflow provides a traceable path from genome QC and target discovery through inclusivity/exclusivity analysis to oligo-level screening, with each design decision documented and reproducible. The panel itself, along with the pipeline scripts that produced it, is committed to the project's GitHub repository. The resulting candidate panel is ready for experimental evaluation, while the underlying workflow can be transferred to target-selection problems in other microbial diagnostic assays.